

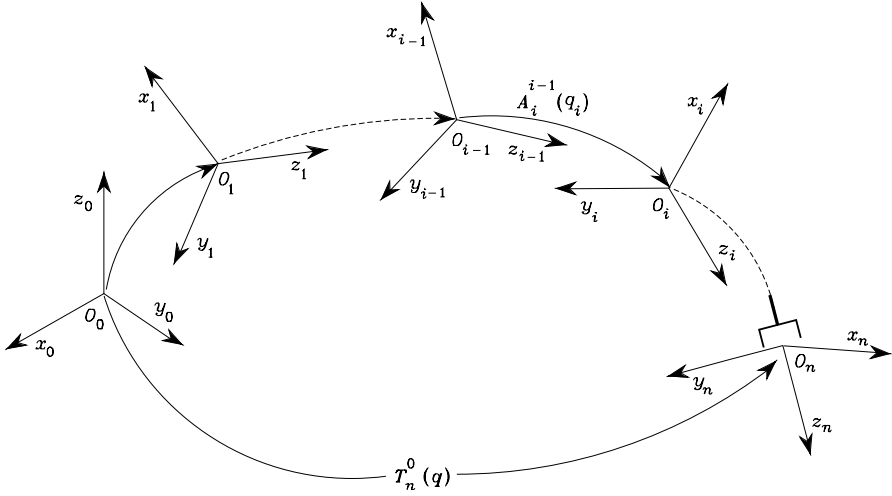


Fig. 2.15. Coordinate transformations in an open kinematic chain

links, it is reasonable to consider first the description of kinematic relationship between consecutive links and then to obtain the overall description of manipulator kinematics in a recursive fashion. To this purpose, it is worth defining a coordinate frame attached to each link, from Link 0 to Link n . Then, the coordinate transformation describing the position and orientation of Frame n with respect to Frame 0 (Fig. 2.15) is given by

$$\mathbf{T}_n^0(\mathbf{q}) = \mathbf{A}_1^0(q_1) \mathbf{A}_2^1(q_2) \dots \mathbf{A}_n^{n-1}(q_n). \quad (2.50)$$

As requested, the computation of the direct kinematics function is recursive and is obtained in a systematic manner by simple products of the homogeneous transformation matrices $\mathbf{A}_i^{i-1}(q_i)$ (for $i = 1, \dots, n$), each of which is a function of a single joint variable.

With reference to the direct kinematics equation in (2.49), the actual coordinate transformation describing the position and orientation of the end-effector frame with respect to the base frame can be obtained as

$$\mathbf{T}_e^b(\mathbf{q}) = \mathbf{T}_0^b \mathbf{T}_n^0(\mathbf{q}) \mathbf{T}_e^n \quad (2.51)$$

where $\mathbf{T}_0^b$ and $\mathbf{T}_e^n$ are two (typically) constant homogeneous transformations describing the position and orientation of Frame 0 with respect to the base frame, and of the end-effector frame with respect to Frame n , respectively.

2.8.2 Denavit–Hartenberg Convention

In order to compute the direct kinematics equation for an open-chain manipulator according to the recursive expression in (2.50), a systematic, general

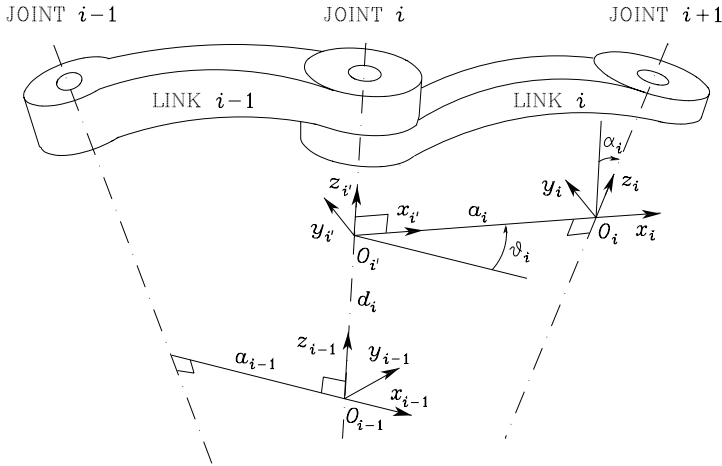


Fig. 2.16. Denavit–Hartenberg kinematic parameters

method is to be derived to define the relative position and orientation of two consecutive links; the problem is that to determine two frames attached to the two links and compute the coordinate transformations between them. In general, the frames can be arbitrarily chosen as long as they are attached to the link they are referred to. Nevertheless, it is convenient to set some rules also for the definition of the link frames.

With reference to Fig. 2.16, let Axis i denote the axis of the joint connecting Link $i - 1$ to Link i ; the so-called *Denavit–Hartenberg convention* (DH) is adopted to define link Frame i :

- Choose axis z_i along the axis of Joint $i + 1$.
- Locate the origin O_i at the intersection of axis z_i with the common normal⁹ to axes z_{i-1} and z_i . Also, locate $O_{i'}$ at the intersection of the common normal with axis z_{i-1} .
- Choose axis x_i along the common normal to axes z_{i-1} and z_i with direction from Joint i to Joint $i + 1$.
- Choose axis y_i so as to complete a right-handed frame.

The Denavit–Hartenberg convention gives a nonunique definition of the link frame in the following cases:

- For Frame 0, only the direction of axis z_0 is specified; then O_0 and x_0 can be arbitrarily chosen.
- For Frame n , since there is no Joint $n + 1$, z_n is not uniquely defined while x_n has to be normal to axis z_{n-1} . Typically, Joint n is revolute, and thus z_n is to be aligned with the direction of z_{n-1} .

⁹ The common normal between two lines is the line containing the minimum distance segment between the two lines.

- When two consecutive axes are parallel, the common normal between them is not uniquely defined.
- When two consecutive axes intersect, the direction of x_i is arbitrary.
- When Joint i is prismatic, the direction of z_{i-1} is arbitrary.

In all such cases, the indeterminacy can be exploited to simplify the procedure; for instance, the axes of consecutive frames can be made parallel.

Once the link frames have been established, the position and orientation of Frame i with respect to Frame $i - 1$ are completely specified by the following *parameters*:

- a_i distance between O_i and $O_{i'}$,
- d_i coordinate of $O_{i'}$ along z_{i-1} ,
- α_i angle between axes z_{i-1} and z_i about axis x_i to be taken positive when rotation is made counter-clockwise,
- ϑ_i angle between axes x_{i-1} and x_i about axis z_{i-1} to be taken positive when rotation is made counter-clockwise.

Two of the four parameters (a_i and α_i) are always constant and depend only on the geometry of connection between consecutive joints established by Link i . Of the remaining two parameters, only one is variable depending on the type of joint that connects Link $i - 1$ to Link i . In particular:

- if Joint i is *revolute* the variable is ϑ_i ,
- if Joint i is *prismatic* the variable is d_i .

At this point, it is possible to express the coordinate transformation between Frame i and Frame $i - 1$ according to the following steps:

- Choose a frame aligned with Frame $i - 1$.
- Translate the chosen frame by d_i along axis z_{i-1} and rotate it by ϑ_i about axis z_{i-1} ; this sequence aligns the current frame with Frame i' and is described by the homogeneous transformation matrix

$$\mathbf{A}_{i'}^{i-1} = \begin{bmatrix} c_{\vartheta_i} & -s_{\vartheta_i} & 0 & 0 \\ s_{\vartheta_i} & c_{\vartheta_i} & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- Translate the frame aligned with Frame i' by a_i along axis $x_{i'}$ and rotate it by α_i about axis $x_{i'}$; this sequence aligns the current frame with Frame i and is described by the homogeneous transformation matrix

$$\mathbf{A}_i^{i'} = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c_{\alpha_i} & -s_{\alpha_i} & 0 \\ 0 & s_{\alpha_i} & c_{\alpha_i} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- The resulting coordinate transformation is obtained by postmultiplication of the single transformations as

$$\mathbf{A}_i^{i-1}(q_i) = \mathbf{A}_{i'}^{i-1} \mathbf{A}_i^{i'} = \begin{bmatrix} c_{\vartheta_i} & -s_{\vartheta_i} c_{\alpha_i} & s_{\vartheta_i} s_{\alpha_i} & a_i c_{\vartheta_i} \\ s_{\vartheta_i} & c_{\vartheta_i} c_{\alpha_i} & -c_{\vartheta_i} s_{\alpha_i} & a_i s_{\vartheta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.52)$$

Notice that the transformation matrix from Frame i to Frame $i-1$ is a function only of the joint variable q_i , that is, ϑ_i for a revolute joint or d_i for a prismatic joint.

To summarize, the Denavit–Hartenberg convention allows the construction of the direct kinematics function by composition of the individual coordinate transformations expressed by (2.52) into one homogeneous transformation matrix as in (2.50). The procedure can be applied to any open kinematic chain and can be easily rewritten in an operating form as follows.

1. Find and number consecutively the joint axes; set the directions of axes $z_0, \dots, z_{n-1}$.
2. Choose Frame 0 by locating the origin on axis z_0 ; axes x_0 and y_0 are chosen so as to obtain a right-handed frame. If feasible, it is worth choosing Frame 0 to coincide with the base frame.

Execute steps from **3** to **5** for $i = 1, \dots, n-1$:

3. Locate the origin O_i at the intersection of z_i with the common normal to axes z_{i-1} and z_i . If axes z_{i-1} and z_i are parallel and Joint i is revolute, then locate O_i so that $d_i = 0$; if Joint i is prismatic, locate O_i at a reference position for the joint range, e.g., a mechanical limit.
4. Choose axis x_i along the common normal to axes z_{i-1} and z_i with direction from Joint i to Joint $i+1$.
5. Choose axis y_i so as to obtain a right-handed frame.

To complete:

6. Choose Frame n ; if Joint n is revolute, then align z_n with z_{n-1} , otherwise, if Joint n is prismatic, then choose z_n arbitrarily. Axis x_n is set according to step 4.
7. For $i = 1, \dots, n$, form the table of parameters $a_i, d_i, \alpha_i, \vartheta_i$.
8. On the basis of the parameters in **7**, compute the homogeneous transformation matrices $\mathbf{A}_i^{i-1}(q_i)$ for $i = 1, \dots, n$.
9. Compute the homogeneous transformation $\mathbf{T}_n^0(\mathbf{q}) = \mathbf{A}_1^0 \dots \mathbf{A}_n^{n-1}$ that yields the position and orientation of Frame n with respect to Frame 0.
10. Given $\mathbf{T}_0^b$ and $\mathbf{T}_e^n$, compute the direct kinematics function as $\mathbf{T}_e^b(\mathbf{q}) = \mathbf{T}_0^b \mathbf{T}_n^0 \mathbf{T}_e^n$ that yields the position and orientation of the end-effector frame with respect to the base frame.

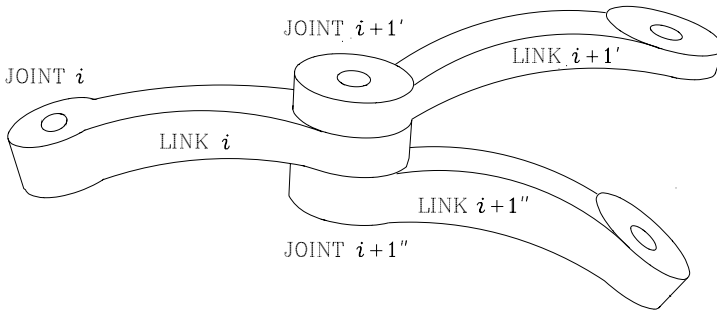


Fig. 2.17. Connection of a single link in the chain with two links

For what concerns the computational aspects of direct kinematics, it can be recognized that the heaviest load derives from the evaluation of transcendental functions. On the other hand, by suitably factorizing the transformation equations and introducing local variables, the number of flops (additions + multiplications) can be reduced. Finally, for computation of orientation it is convenient to evaluate the two unit vectors of the end-effector frame of simplest expression and derive the third one by vector product of the first two.

2.8.3 Closed Chain

The above direct kinematics method based on the DH convention exploits the inherently recursive feature of an open-chain manipulator. Nevertheless, the method can be extended to the case of manipulators containing closed kinematic chains according to the technique illustrated below.

Consider a *closed-chain* manipulator constituted by $n + 1$ links. Because of the presence of a loop, the number of joints l must be greater than n ; in particular, it can be understood that the number of closed loops is equal to $l - n$.

With reference to Fig. 2.17, Links 0 through i are connected successively through the first i joints as in an open kinematic chain. Then, Joint $i + 1'$ connects Link i with Link $i + 1'$ while Joint $i + 1''$ connects Link i with Link $i + 1''$; the axes of Joints $i + 1'$ and $i + 1''$ are assumed to be aligned. Although not represented in the figure, Links $i + 1'$ and $i + 1''$ are members of the closed kinematic chain. In particular, Link $i + 1'$ is further connected to Link $i + 2'$ via Joint $i + 2'$ and so forth, until Link j via Joint j . Likewise, Link $i + 1''$ is further connected to Link $i + 2''$ via Joint $i + 2''$ and so forth, until Link k via Joint k . Finally, Links j and k are connected together at Joint $j + 1$ to form a closed chain. In general, $j \neq k$.

In order to attach frames to the various links and apply DH convention, one closed kinematic chain is taken into account. The closed chain can be virtually cut open at Joint $j + 1$, i.e., the joint between Link j and Link k . An equivalent tree-structured open kinematic chain is obtained, and thus link